1. **Command Name: ‘mkdir’**

Purpose:

- ‘mkdir’ means make directory.
- It is used to create new folder on current directory.
- Syntax- *mkdir folder_name*

```
adil@ASUS-Vivobook-Intel-corei7:~$ ls
adil@ASUS-Vivobook-Intel-corei7:~$ |
```

(Without making directory)

```
adil@ASUS-Vivobook-Intel-corei7:~$ mkdir dummy1
adil@ASUS-Vivobook-Intel-corei7:~$ ls
dummy1
adil@ASUS-Vivobook-Intel-corei7:~$ |
```

(After making directory)

Used flag: **-p** → which is used to create nested folder.

```
adil@ASUS-Vivobook-Intel-corei7:~$ mkdir -p dummy1/dummy2
adil@ASUS-Vivobook-Intel-corei7:~$ ls
dummy1
adil@ASUS-Vivobook-Intel-corei7:~$ cd dummy1
adil@ASUS-Vivobook-Intel-corei7:~/dummy1$ ls
dummy2
adil@ASUS-Vivobook-Intel-corei7:~/dummy1$ |
```

Used flag: -m → it helps to create folder with custom permissions.

```
adil@ASUS-Vivobook-Intel-corei7:~$ mkdir -m 755 mode.txt
adil@ASUS-Vivobook-Intel-corei7:~$ ls -ld
drwxr-x--- 7 adil adil 4096 Sep 21 11:38 .
adil@ASUS-Vivobook-Intel-corei7:~$ ls -ld mode.txt
drwxr-xr-x 2 adil adil 4096 Sep 21 11:38 mode.txt
adil@ASUS-Vivobook-Intel-corei7:~$
```

Used flag: -v → it returns a confirmation message after creating directory.

```
adil@ASUS-Vivobook-Intel-corei7:~$ mkdir -v new
mkdir: created directory 'new'
adil@ASUS-Vivobook-Intel-corei7:~$ |
```

```
adil@ASUS-Vivobook-Intel-corei7:~$ ls
copy.txt   dummy1   file.txt   mode.txt   new
adil@ASUS-Vivobook-Intel-corei7:~$ |
```

2. **Command name: ‘touch’**

Purpose:

- It is used to create different kinds of files with extensions.
- Syntax- *touch file_name.extension*

```
adil@ASUS-Vivobook-Intel-corei7:~$ ls
dummy1
adil@ASUS-Vivobook-Intel-corei7:~$ touch text.txt
adil@ASUS-Vivobook-Intel-corei7:~$ ls
dummy1  text.txt
adil@ASUS-Vivobook-Intel-corei7:~$
```

Used flag: -c → it changes the timestamp if the file exists only.

```
adil@ASUS-Vivobook-Intel-corei7:~$ touch -c new.txt
adil@ASUS-Vivobook-Intel-corei7:~$ ls
copy.txt  dummy1  file.txt  mode.txt  new
adil@ASUS-Vivobook-Intel-corei7:~$ ls new.txt
ls: cannot access 'new.txt': No such file or directory
adil@ASUS-Vivobook-Intel-corei7:~$
```

Used flag: -c → it helps to specify the custom timestamp of the file.

```
adil@ASUS-Vivobook-Intel-corei7:~$ touch -d "2025-01-01 10:30:
00" file2.txt
adil@ASUS-Vivobook-Intel-corei7:~$ ls
copy.txt  dummy1  file.txt  file2.txt  mode.txt  new
adil@ASUS-Vivobook-Intel-corei7:~$ stat file2.txt
  File: file2.txt
  Size: 0               Blocks: 0          IO Block: 4096   re
gular empty file
Device: 8,48    Inode: 2595        Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/    adil)   Gid: ( 1000
/    adil)
Access: 2025-01-01 10:30:00.000000000 +0600
Modify: 2025-01-01 10:30:00.000000000 +0600
Change: 2025-09-21 11:50:44.323254775 +0600
 Birth: 2025-09-21 11:50:44.323254775 +0600
adil@ASUS-Vivobook-Intel-corei7:~$
```

### 3. Command name: 'ls'

Purpose:

- It is used to show the files, folders of current directory.

```
adil@ASUS-Vivobook-Intel-corei7:~$ cd dummy1
adil@ASUS-Vivobook-Intel-corei7:~/dummy1$ ls
dummy2
```

Used flag: **-a** → to show the hidden files and folders of current directory.

```
adil@ASUS-Vivobook-Intel-corei7:~$ ls
dummy1  text.txt
adil@ASUS-Vivobook-Intel-corei7:~$ ls -a
.              .bashrc     .local                      dummy1
..             .cache      .motd_shown                 text.txt
.bash_history  .landscape  .profile
.bash_logout   .lesshst    .sudo_as_admin_successful
adil@ASUS-Vivobook-Intel-corei7:~$
```

Used flag: **-f** → to show all the list in directory order.

```
adil@ASUS-Vivobook-Intel-corei7:~$ ls -f
.profile                    file2.txt      .bash_logout
..                          .bash_history  .cache
copy.txt                    .motd_shown    .lesshst
.                           .bashrc        dummy1
.sudo_as_admin_successful   .landscape     .local
new                         mode.txt       file.txt
adil@ASUS-Vivobook-Intel-corei7:~$
```

Used flag: **-l**→ show all the information of the files and folders of current directory.

```
adil@ASUS-Vivobook-Intel-corei7:~$ ls -l
total 20
-rw-r--r-- 1 adil adil   17 Sep 20 17:59 copy.txt
drwxr-xr-x 2 adil adil 4096 Sep 20 17:42 dummy1
-rw-r--r-- 1 adil adil   17 Sep 20 17:48 file.txt
-rw-r--r-- 1 adil adil    0 Jan  1  2025 file2.txt
drwxr-xr-x 2 adil adil 4096 Sep 21 11:38 mode.txt
drwxr-xr-x 2 adil adil 4096 Sep 21 11:42 new
adil@ASUS-Vivobook-Intel-corei7:~$
```

4. **Command Name: ‘pwd’**

Purpose:

- It is used to show the current directory address.

```
adil@ASUS-Vivobook-Intel-corei7:~$ pwd
/home/adil
adil@ASUS-Vivobook-Intel-corei7:~$ cd dummy1
adil@ASUS-Vivobook-Intel-corei7:~/dummy1$ pwd
/home/adil/dummy1
adil@ASUS-Vivobook-Intel-corei7:~/dummy1$
```

Used flag: **-P**⟶ It helps to print the real, physical directory on disk by resolving any symbolic links.

```
adil@ASUS-Vivobook-Intel-corei7:~$ pwd -P
/home/adil
adil@ASUS-Vivobook-Intel-corei7:~$
```

Used flag: **-L**⟶ It helps to print the logical path which is default.

```
adil@ASUS-Vivobook-Intel-corei7:~$ pwd -L
/home/adil
adil@ASUS-Vivobook-Intel-corei7:~$
```

5. **Command name: ‘rm’**

Purpose:

- It is used to remove or delete a file
- Syntax- rm *file_name*

```
adil@ASUS-Vivobook-Intel-corei7:~$ ls
dummy1  text.txt
adil@ASUS-Vivobook-Intel-corei7:~$ rm text.txt
adil@ASUS-Vivobook-Intel-corei7:~$ ls
dummy1
adil@ASUS-Vivobook-Intel-corei7:~$
```

Used flag: **-r** ⟶ used to delete a folder.

```
adil@ASUS-Vivobook-Intel-corei7:~/dummy1$ ls
dummy2
adil@ASUS-Vivobook-Intel-corei7:~/dummy1$ rm dummy2
rm: cannot remove 'dummy2': Is a directory
adil@ASUS-Vivobook-Intel-corei7:~/dummy1$ rm -r dummy2
adil@ASUS-Vivobook-Intel-corei7:~/dummy1$ ls
adil@ASUS-Vivobook-Intel-corei7:~/dummy1$
```

Used flag: **-i** ⟶ It asks before delete a file.

```
adil@ASUS-Vivobook-Intel-corei7:~$ rm -i file2.txt
rm: remove regular empty file 'file2.txt'? y
adil@ASUS-Vivobook-Intel-corei7:~$ ls
copy.txt  dummy1  file.txt  mode.txt  new
adil@ASUS-Vivobook-Intel-corei7:~$
```

Used flag: **-f** ⟶ It does not give confirmation and no error if file does not exist.

```
adil@ASUS-Vivobook-Intel-corei7:~$ rm file2.txt
rm: cannot remove 'file2.txt': No such file or directory
adil@ASUS-Vivobook-Intel-corei7:~$ rm -f file2.txt
adil@ASUS-Vivobook-Intel-corei7:~$
```

6. **Command name: 'cat'**

Purpose:

- cat means concatenate
- It is used to display the contents of the file.
- Syntax- *cat file_name*

```
adil@ASUS-Vivobook-Intel-corei7:~$ touch file.txt
adil@ASUS-Vivobook-Intel-corei7:~$ echo Operating System>file.txt
adil@ASUS-Vivobook-Intel-corei7:~$ cat file.txt
Operating System
adil@ASUS-Vivobook-Intel-corei7:~$
```

Used flag: **-n** ⟶ which is used to show the output with line numbers

```
adil@ASUS-Vivobook-Intel-corei7:~$ cat -n file.txt
     1  Operating System
adil@ASUS-Vivobook-Intel-corei7:~$
```

Used flag: **-E** ⟶ it shows $ sign end of every line.

```
adil@ASUS-Vivobook-Intel-corei7:~$ cat -E copy.txt
Operating System$
adil@ASUS-Vivobook-Intel-corei7:~$
```

7. **Command name: ‘cp’**

Purpose:

- It is used to copy a file with different name in same directory
- Syntax- *cp main_file_name copy_file_name*

```
adil@ASUS-Vivobook-Intel-corei7:~$ ls
dummy1  file.txt
adil@ASUS-Vivobook-Intel-corei7:~$ cp file.txt copy.txt
adil@ASUS-Vivobook-Intel-corei7:~$ ls
copy.txt  dummy1  file.txt
adil@ASUS-Vivobook-Intel-corei7:~$ |
```

Used flag: **-i** ⟶ It asks before overwrite the file.

```
adil@ASUS-Vivobook-Intel-corei7:~$ cp -i copy.txt file.txt
cp: overwrite 'file.txt'? y
adil@ASUS-Vivobook-Intel-corei7:~$ |
```

Used flag: **-v**⟶ It shows what is being copied.

```
adil@ASUS-Vivobook-Intel-corei7:~$ cp -v copy.txt file.txt
'copy.txt' -> 'file.txt'
adil@ASUS-Vivobook-Intel-corei7:~$ |
```

8. **Command name: ‘grep’**

Purpose:

- It is used to search for words in the files
- Syntax- *grep “word” file_name*

```
adil@ASUS-Vivobook-Intel-corei7:~$ grep System file.txt
Operating System
adil@ASUS-Vivobook-Intel-corei7:~$ |
```

```
adil@ASUS-Vivobook-Intel-corei7:~$ grep System *
copy.txt:Operating System
grep: dummy1: Is a directory
file.txt:Operating System
adil@ASUS-Vivobook-Intel-corei7:~$ |
```

Used flag: **-i** ⟶ which helps to ignore the case sensitive problem

```
adil@ASUS-Vivobook-Intel-corei7:~$ grep -i system file.txt
Operating System
adil@ASUS-Vivobook-Intel-corei7:~$ |
```

Used flag: **-c** ⟶ which shows the total number of exact matching word, it is case sensitive

```
adil@ASUS-Vivobook-Intel-corei7:~$ grep -c system file.txt
0
adil@ASUS-Vivobook-Intel-corei7:~$ grep -c System file.txt
1
adil@ASUS-Vivobook-Intel-corei7:~$
```

Used flag: **-n**⟶ which shows the texts with line number.

```
adil@ASUS-Vivobook-Intel-corei7:~$ grep -n System file.txt
1:Operating System
adil@ASUS-Vivobook-Intel-corei7:~$
```

Used flag: **-v**⟶ which shows the non-matching lines.

```
adil@ASUS-Vivobook-Intel-corei7:~$ cat file.txt
Operating System
CSE 3202 Lab Sessional
adil@ASUS-Vivobook-Intel-corei7:~$ grep -v System file.txt
CSE 3202 Lab Sessional
adil@ASUS-Vivobook-Intel-corei7:~$
```